Optimal Light Spectra Under Regulatory Constraints: How Efficiency Standards Drive Lamps to Weird Spectral States

Generated by `photopic` — a Modula-3/Scheme spectral optimizer

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Abstract

Government regulations on lamp efficiency (lumens per watt) and color quality (Color Rendering Index) define a constrained optimization problem whose solutions are not the smooth, broad spectra of incandescent bulbs but narrow-band, spiky spectral power distributions. We use `photopic`, a hybrid Modula-3/Scheme numerical tool, to explore the Pareto frontier of this trade-off. As the efficacy requirement increases, the optimal spectrum converges to approximately three narrow peaks near 450, 540, and 610 nm — Thornton's "prime color" wavelengths — while aggressively suppressing power at all other wavelengths, including the red region critical for color fidelity. The results show that regulatory pressure, when pushed to its logical extreme, drives light sources toward spectra that are visually "weird": they satisfy the letter of color rendering standards while producing light that many people find unnatural.

1 Introduction

An incandescent lamp is a near-perfect blackbody radiator. Its spectrum is broad and smooth, rendering colors with perfect fidelity (CRI = 100). But most of its radiation falls in the infrared, giving it a luminous efficacy of only ∼15 lm/W. Government regulations worldwide — EU Ecodesign, US EISA/DOE, California Title 24 — have mandated minimum efficacy standards that incandescent lamps cannot meet, effectively banning them.

The replacement technologies (fluorescent, LED) achieve high efficacy by concentrating their emission in the visible spectrum. But *where* in the visible spectrum? The CIE photopic luminous efficiency function $V(\lambda)$ peaks sharply at 555 nm (yellow-green). A monochromatic source at 555 nm would achieve the theoretical maximum of 683 lm/W — but it would render no colors at all.

The question becomes: **What spectral power distribution maximizes luminous efficacy while maintaining acceptable color rendering?**

This is a constrained optimization problem. The constraints come from regulations:

- **CRI $\geq$ 80** (Energy Star, EU); CRI $\geq$ 90 (California Title 24 for some applications)
- **CCT within ±50K** of the nominal color temperature (e.g., 2700K for "warm white")
- **Duv $\leq$ 0.012**: chromaticity tolerance (distance from the blackbody locus)

We explore what happens as you "turn up the knob" on the efficacy requirement. The answer is surprising: the optimal spectra become increasingly bizarre.

2 Background

2.1 Luminous Efficacy

Luminous efficacy of radiation (LER) is the ratio of luminous flux to radiant power:

$$\eta = K_m \frac{\int_0^\infty S(\lambda)\, V(\lambda)\, d\lambda}{\int_0^\infty S(\lambda)\, d\lambda} \tag{1}$$

where $S(\lambda)$ is the spectral power distribution, $V(\lambda)$ is the CIE 1931 photopic luminous efficiency function, and $K_m = 683$ lm/W is the maximum luminous efficacy at 555 nm.

For a 2700K blackbody, the total LER is only ∼15 lm/W because most radiation is in the infrared. If we restrict to the visible band (380–770 nm), the visible-only LER is ∼88 lm/W.

2.2 Color Rendering Index (CRI)

The CRI quantifies how faithfully a light source renders colors compared to a reference illuminant (a blackbody at the same correlated color temperature). The calculation, following CIE 13.3-1995 and the colorimetric formulas in Foley and van Dam [5], involves:

1. Normalize the test spectrum to $Y = 100$
2. Find the reference blackbody at the correlated color temperature (CCT) via search in CIE 1960 MacAdam (u, v) space
3. For each of 14 Test Color Samples (TCS), compute the color shift ΔE^*_{UVW} using von Kries chromatic adaptation
4. $R_i = 100 - 4.6\, \Delta E^*_{UVW}$ for each sample; $\mathrm{CRI}(R_a) = \frac{1}{8} \sum_{i=1}^{8} R_i$

The key subtlety: only TCS 1–8 contribute to R_a. TCS 9 (saturated red) is *not* part of the official CRI score. This creates a loophole that the optimizer exploits.

2.3 Thornton's Prime Colors

In 1971, W. A. Thornton of Westinghouse showed that the most chromatically effective wavelengths for white light are three narrow bands near **450, 540, and 610 nm** — the "prime colors" [1]. Conversely, wavelengths near 500 nm and 580 nm are "anti-prime" — they contribute power without adding chromaticity discrimination. Thornton was named US National "Inventor of the Year" in 1979 for his prime-color fluorescent lamp patents [4].

The optimization results below independently rediscover Thornton's prime colors from first principles.

3 The `photopic` Tool

`photopic` is a hybrid application: performance-critical spectral calculations (Planck's law, CIE color matching functions, numerical integration) are implemented in compiled Modula-3 [11, 12] libraries, while the optimization driver, CRI calculation, and plotting are written in Scheme using MScheme [13], a Scheme interpreter embedded in the Modula-3 runtime. MScheme's `sstubgen` tool automatically generates Scheme bindings from Modula-3 interfaces, allowing the Scheme code

to manipulate compiled M3 objects (spectral arrays, integration routines, color-matching data) directly without marshalling overhead. This architecture provides the rapid prototyping benefits of a scripting language for the optimization logic while retaining compiled-language performance for the inner numerical loops.

3.1 Architecture

Component	Role
`Blackbody.m3`	Planck radiance $B_\nu(T, \nu)$
`CieSpectrum.m3`	Photopic $V(\lambda)$, conversion to lumens
`CieXyz.m3`	CIE 1931 $\bar{x}, \bar{y}, \bar{z}$ color matching
`Tcs.m3`	14 Test Color Sample reflectance spectra
`TempUv.m3`	Blackbody locus in (u, v) for CCT search
`ParametricSpectrum.m3`	Piecewise-linear log-amplitude parameterization
`COBYLA_M3.m3`	Constrained optimization (Powell's COBYLA)
`NR4p4.m3`	Romberg integration (midpoint rule)
`Bracket.m3`	Brent's method for CCT root-finding
`photopic.scm`	Optimization driver, CRI calc, plotting

3.2 Spectrum Parameterization

The optimizer works in log-amplitude space. A test spectrum is represented as:

$$S(\lambda) = \exp\big(f(\lambda)\big) \cdot B_\lambda(T_0, \lambda) \tag{2}$$

where $B_\lambda(T_0, \lambda)$ is the baseline blackbody at the target CCT, and $f(\lambda)$ is a piecewise-linear function defined by n parameters (control points evenly spaced across 380–770 nm).

The resolution increases through *hierarchical subdivision*: $n = 2 \to 3 \to 5 \to 9 \to 17 \to 33 \to 65 \to 129$. At each level, COBYLA [6] minimizes the negative efficacy subject to CRI, CCT, and chromaticity constraints. The solution from level n is interpolated to initialize level $2n - 1$.

3.3 Objective Function

The optimization target vector passed to COBYLA is:

$$\mathbf{t} = \begin{pmatrix} -\eta \\ R_a - R_a^{\min} \\ R_9 - R_9^{\min} \\ R_{\min} - (R_a^{\min} - 10) \\ \mathrm{CCT} - (T_0 - 50) \\ (T_0 + 50) - \mathrm{CCT} \\ 1000\,(0.012 - D_{uv}) \end{pmatrix} \tag{3}$$

where η is the LER, and the remaining components are inequality constraints (≥ 0 required).

4 Results

4.1 Blackbody vs. Optimized Spectrum

Figure 1 shows the dramatic difference between a 2700K blackbody and the maximum-efficacy spectrum with CRI ≥ 60.

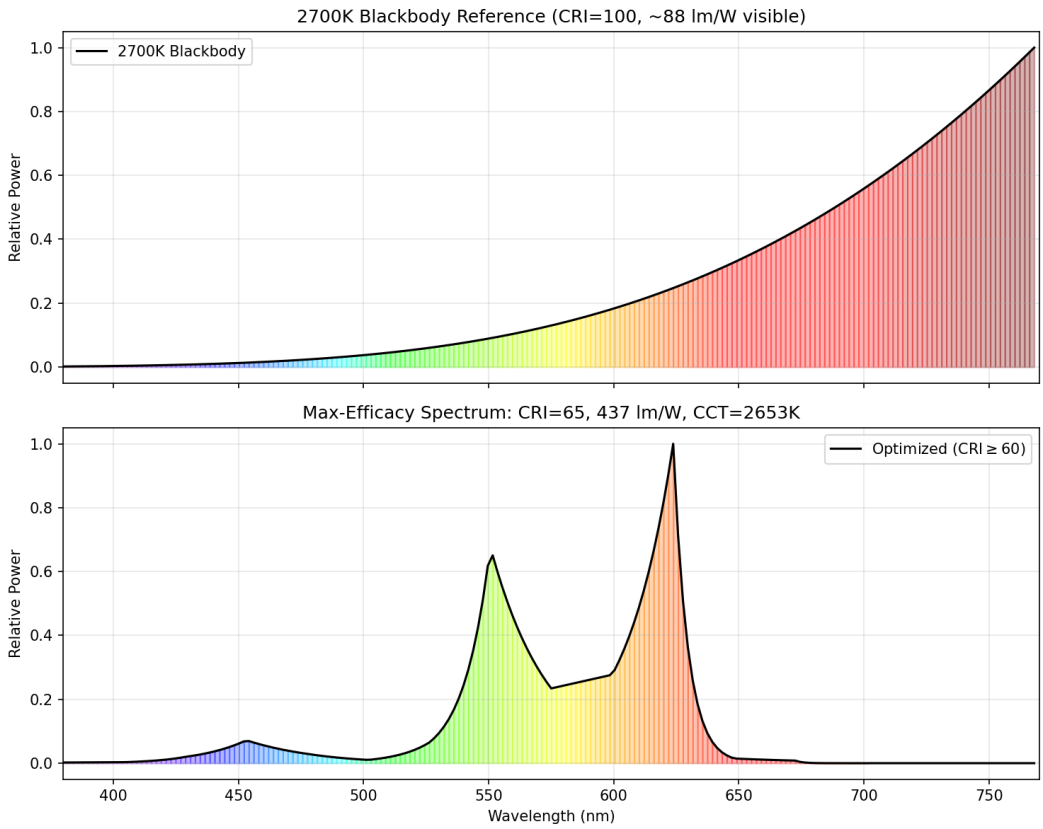

Figure 1: Top: 2700K blackbody reference spectrum (CRI = 100, ∼88 lm/W visible). Bottom: maximum-efficacy spectrum achieving CRI = 65 at 437 lm/W. The optimizer concentrates power into three peaks near Thornton's prime-color wavelengths while suppressing power below 500 nm and above 660 nm.

The optimized spectrum achieves ∼5× higher efficacy than the blackbody by eliminating "wasted" power at wavelengths where the eye is insensitive or where chromaticity is redundant. The three peaks correspond to Thornton's prime colors:

- **∼450 nm** (blue): establishes the short-wavelength chromaticity anchor
- **∼540 nm** (green): near the peak of $V(\lambda)$ — provides most of the lumens
- **∼610 nm** (orange-red): establishes warm color temperature

4.2 Convergence: The Spectrum Gets "Weirder" with More Freedom

Figure 2 shows how the optimized spectrum evolves as the number of control parameters increases from 2 to 17.

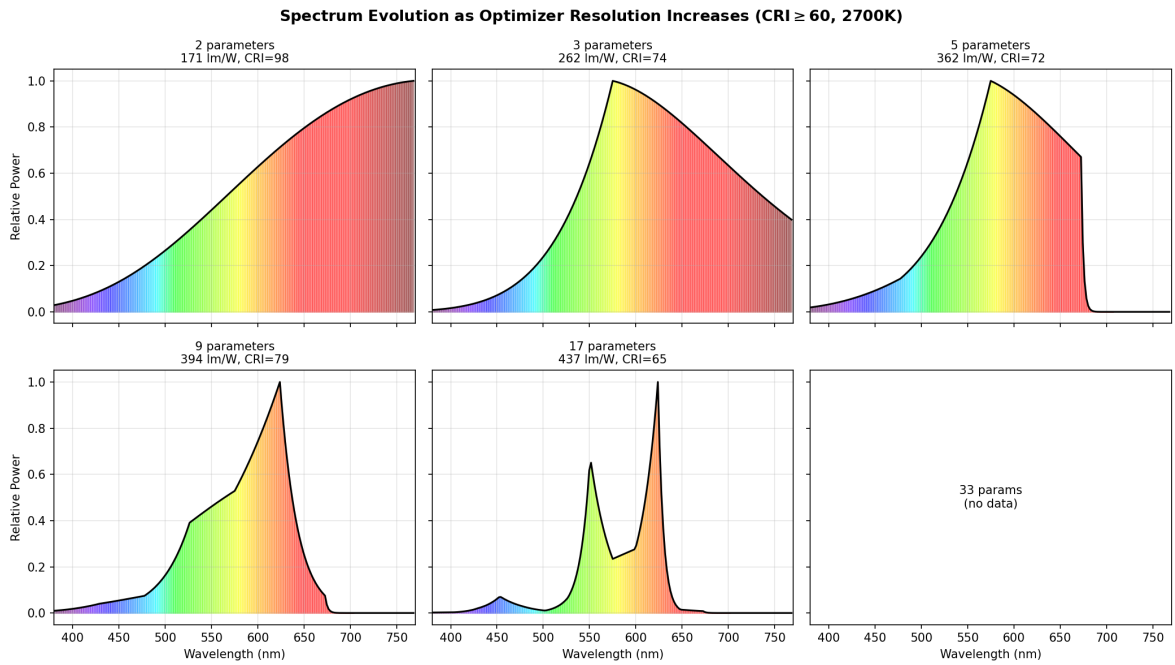

Figure 2: Evolution of the optimal CRI $\geq$ 60 spectrum at 2700K as the optimizer gains more spectral resolution. With 2 parameters, the spectrum is nearly a tilted blackbody (171 lm/W). As parameters increase, the optimizer carves out deep nulls between peaks, reaching 437 lm/W at 17 parameters. The three-peak structure emerges clearly by 9 parameters.

The progression is illuminating:

- **2 parameters** (171 lm/W): Can only tilt the spectrum. Shifts power toward green. Still smooth, CRI = 98.
- **3 parameters** (262 lm/W): First sign of spectral sculpting. Blue and deep red suppressed.
- **5 parameters** (362 lm/W): Three-peak structure begins to emerge. Deep null around 500 nm.
- **9 parameters** (394 lm/W): Clear three peaks with valleys between. CRI drops to 79.
- **17 parameters** (437 lm/W): Fully developed spiky spectrum. Deep nulls, sharp peaks. CRI just satisfies the $\geq$60 floor.

The optimizer, given enough degrees of freedom, always converges to a three-peak solution. **This is Thornton's prime-color theorem, rediscovered numerically.**

4.3 The R9 Problem

The CRI score R_a averages only TCS 1–8, which are relatively unsaturated. TCS 9 (saturated red) is reported separately. When the optimizer is free to set R_9 as low as -100, it exploits this: the CRI $\geq$ 60 optimized spectrum at 17 parameters has $R_9 = -28$.

This means: under this "optimal" light, saturated reds look dramatically wrong — less red than under a reference illuminant. A person wearing a red dress would look dull; red paint would look brownish. But the CRI score says the light is acceptable.

This is the "weird final state" that regulations drive toward: a lamp that passes the official color test while rendering an important class of colors — reds — poorly.

4.4 Effect of CRI Constraints

Figure 3 shows how the optimal spectrum changes as the CRI floor increases.

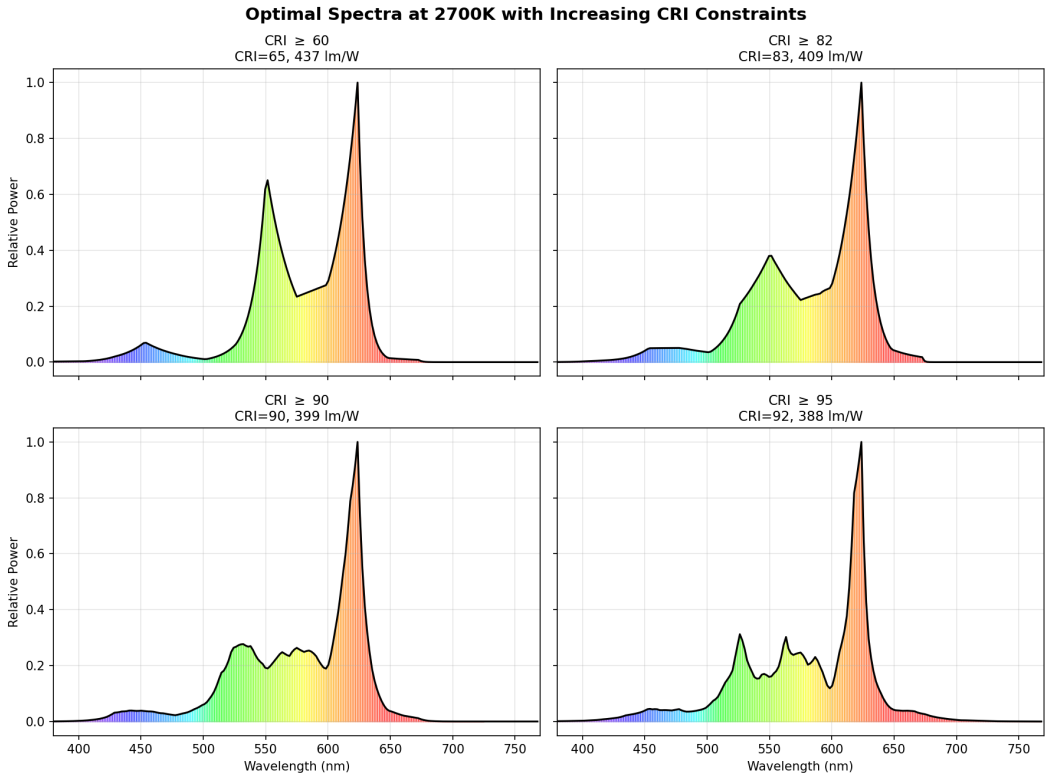

Figure 3: Optimal spectra at 2700K for different CRI constraints. As CRI requirements increase, the spectrum must broaden to render more colors faithfully, reducing the maximum achievable efficacy. The three-peak structure persists but the peaks must widen and the valleys cannot be as deep.

Table 1: Efficacy–CRI trade-off at 2700K CCT, unconstrained R_9. Results at 129 spectral parameters (17 for CRI 60).

Min CRI	Achieved CRI	Efficacy (lm/W)	R_9	Worst R_i	Character
—	100	88	100	100	2700K blackbody (visible only)
60	65	437	−28	50	Three sharp peaks, deep valleys
70	73	428	−7	60	Three peaks, moderate valleys
80	83	412	+21	70	EU minimum: R9 still poor
82	83	409	+29	72	Three peaks, shallower valleys
85	88	402	+33	75	Transitional region
90	90	399	+45	80	Broader peaks, less suppression
95	92	388	+58	86	Wide peaks, modest valleys
98	93	358	+75	89	Approaching continuous

The trade-off is dramatic: going from CRI 60 to CRI 98 costs roughly 80 lm/W of achievable efficacy (437 → 358 lm/W at 129 spectral parameters), and the key variable is R_9 — the ability to render saturated reds. At CRI 60, R_9 is deeply negative (−28), meaning reds appear wrong; at CRI 98, $R_9 = 75$ and red rendering is acceptable.

The EU regulatory minimum — CRI ≥ 80 at ∼412 lm/W — achieves high efficiency but with $R_9 = 21$, meaning reds are rendered at barely one-fifth of reference fidelity. This is the "weird final state" that market and regulatory forces converge on. California's JA8 standard, which requires $R_9 \geq 50$, pushes manufacturers above CRI 90, where the achievable efficacy drops to ∼399 lm/W but red rendering reaches $R_9 = 45$.

4.5 The Pareto Frontier: Overlay of All Solutions

Figure 4 overlays all optimized spectra on a single plot, together with the blackbody reference.

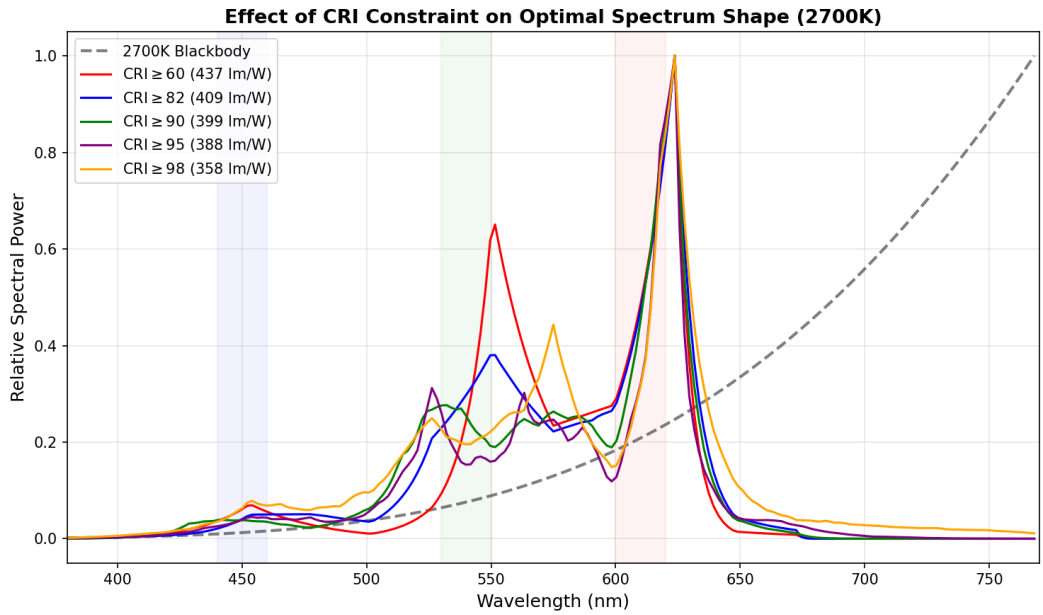

Figure 4: Overlay of optimal spectra at 2700K for CRI constraints from 60 to 98. All spectra share the same three-peak topology (blue ∼450 nm, green ∼540 nm, red ∼610 nm) but differ in peak width and valley depth. The dashed curve is the 2700K blackbody. The shaded bands mark Thornton's prime-color wavelength regions.

4.6 The Efficacy–CRI Trade-off Curve

Figure 5 plots the maximum achievable efficacy as a function of the CRI floor.

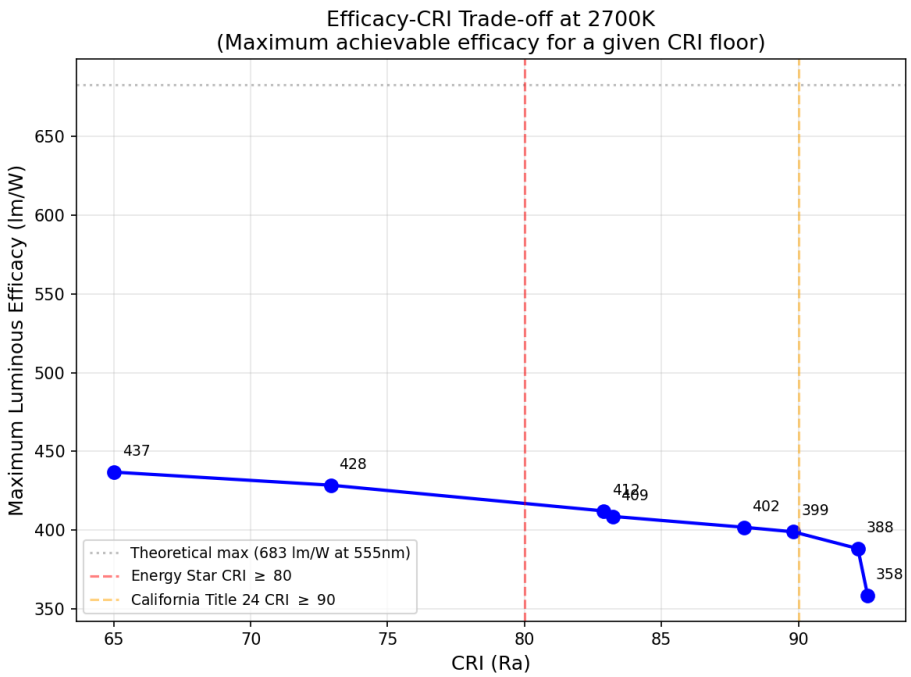

Figure 5: Pareto frontier: maximum luminous efficacy vs. minimum CRI at 2700K. Vertical dashed lines show the Energy Star (CRI $\geq$ 80) and California Title 24 (CRI $\geq$ 90) thresholds. The theoretical maximum at 555 nm is 683 lm/W. The curve shows rapidly diminishing returns: moving from CRI 82 to CRI 98 costs about 80 lm/W.

4.7 R_9 vs. Efficacy: The Red Rendering Penalty

Figure 6 shows the key result: at the EU minimum CRI $\geq$ 80, the optimal spectrum achieves 412 lm/W theoretical luminous efficacy of radiation (LER) but with $R_9 = 21$. Saturated reds are rendered at only one-fifth of their reference fidelity. To achieve California JA8's $R_9 \geq 50$ requirement, CRI must be pushed above 90, reducing achievable efficacy to $\sim$399 lm/W.

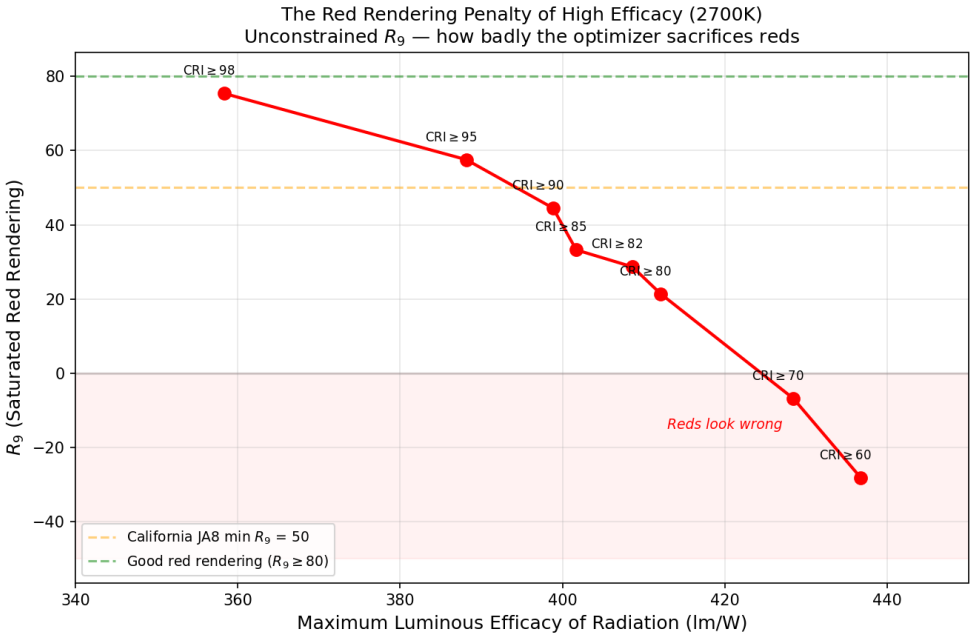

Figure 6: R_9 (saturated red rendering index) as a function of maximum achievable luminous efficacy of radiation at 2700K. Each point represents the optimal spectrum at a different CRI floor, with R_9 unconstrained. At the EU minimum (CRI $\geq$ 80), R_9 is only 21. The California JA8 minimum of $R_9 \geq 50$ is not met until CRI $\geq$ 90.

These are theoretical radiation-only efficacies assuming unit quantum efficiency. For real phosphor-converted LED lamps, Koedam and Opstelten estimated a ∼20% reduction due to realistic quantum efficiency and Stokes losses [10]. The correction is wavelength-dependent: converting a 450 nm blue pump photon to a longer-wavelength visible photon loses a fraction $1-450/\lambda_{\text{em}}$ of the photon energy as heat (the Stokes shift). Table 2 shows the penalty for representative LED phosphors.

Table 2: Stokes loss and quantum yield for LED phosphors (450 nm blue pump).

Phosphor	Emission color	λ_{em} (nm)	Stokes loss	Typ. QY	Net loss
YAG:Ce	yellow	560	19.6%	96%	23%
β-SiAlON:Eu	green	535	15.9%	83%	30%
KSF (K_2SiF_6:Mn)	red	631	28.7%	95%	33%
Nitride (CASN)	deep red	650	30.8%	88%	39%

Adding red phosphor content — needed for good R_9 — incurs a *double* penalty: (1) the Stokes loss is higher (29–31% vs. 16–20% for green/yellow), and (2) the eye's photopic sensitivity $V(\lambda)$ drops steeply in the red, so each red photon contributes fewer lumens. This is exactly why the optimizer suppresses red emission when R_9 is unconstrained, and why achieving good red rendering in real LED lamps is expensive in terms of system efficacy.

5 Discussion

5.1 The Regulatory Paradox

The results reveal a paradox in lighting regulation:

1. Regulations set a *minimum CRI* (typically 80) and a *minimum efficacy* (80–100 lm/W and rising).
2. The optimal solution for high efficacy at moderate CRI is a narrow-band, spiky spectrum.
3. Spiky spectra can achieve high CRI scores (by construction — the 8 test colors are rendered well) while having poor real-world color rendering (especially saturated colors, TCS 9–14).
4. As efficacy requirements increase over time, manufacturers are pushed further toward spiky spectra.
5. The "efficient" lamp of the future satisfies all regulations but produces light that looks *wrong* to many observers.

This is not a hypothetical concern. Commercial tri-phosphor fluorescent lamps and phosphor-converted LEDs already exploit exactly this trade-off. Their spectral power distributions, with 3–5 peaks and deep valleys, are the market's practical approximation to the mathematically optimal solution computed here.

Table 3 compares the major regulatory frameworks. The EU sets CRI ≥ 80 with no R_9 requirement; our optimizer shows that this permits $R_9 = 21$ (Table 1). The US DOE 2028 standard requires ∼125 lm/W system efficacy but imposes no color quality floor at all. Only California JA8 mandates $R_9 \geq 50$ together with CRI ≥ 90.

Table 3: Regulatory landscape for general-service lamps (2025–2026).

	EU 2019/2020	US DOE 2022	US DOE 2028	CA Title 24 JA8
Min system efficacy*	∼98	45	125	45
Min CRI (R_a)	80	—	—	90
Min R_9	—	—	—	50
CCT range	any	any	any	2700–4000 K

*lm/W for an 800 lm non-directional lamp.

5.2 Comparison with Thornton's Work

Thornton arrived at his prime-color wavelengths (450, 540, 610 nm) through a combination of psychophysical experiments and engineering intuition in the 1970s [1, 2, 3]. The `photopic` optimizer, starting from first principles (Planck's law, CIE color matching functions, von Kries adaptation), converges to the same wavelengths. This provides independent numerical confirmation of Thornton's insight.

The deeper significance is that the prime-color wavelengths are not arbitrary — they are the eigenvectors of the human visual system's color discrimination capacity, weighted by luminous sensitivity.

5.3 What the Regulations Should Do

The IES TM-30-15 standard addresses some of these problems by using 99 test color samples instead of 8, and reporting both *fidelity* and *gamut area*. Mandating minimum R_9 scores (as some California specifications do) partially closes the red-rendering loophole. But the fundamental tension between efficacy and color quality is a physical constraint that no regulation can eliminate — it can only choose where on the Pareto frontier to operate.

6 Literature Review

The optimization of white-light spectra has a history reaching back to the 1950s, but two papers from the 1970s are particularly relevant to the present work: Thornton's 1971 study of prime colors and Walter's 1978 application of nonlinear programming to lamp spectra.

6.1 Thornton (1971): Prime Colors and the Structure of White Light

W. A. Thornton of Westinghouse [1] posed the question: *What is the optimum spectral power distribution of artificial white light?* Starting from the CIE luminosity function $\bar{y}(\lambda)$ and the IES–CIE color rendering index, Thornton evaluated white light composed of three spectral lines. By systematic variation of each wavelength while holding the other two fixed (at 450, 540, and 610 nm), he showed that these three wavelengths are local optima of both CRI and a combined figure of merit $\mathrm{FM} = 320\eta + \mathrm{CRI}$.

Thornton's key findings:

- Three spectral lines at **450, 540, and 610 nm** — the "prime colors" — yield white light superior to commercial fluorescent lamps of the same chromaticity in *both* luminosity and CRI. Three-line light at 4200 K achieved $\eta = 0.58$, CRI = 71, compared to $\eta = 0.48$, CRI = 67 for a commercial cool-white fluorescent.

- Wavelengths near **495 and 575 nm** are "anti-prime": they degrade CRI without contributing significantly to luminosity. Removing a slice of power near 500 or 580 nm from a cool-white lamp's spectrum actually *raises* the CRI.

- A 1966 demonstration with interference filters and projectors showed that 60 observers judged three-line prime-color light as providing "very good, if not excellent" color rendering of meat, vegetables, flowers, and skin tones.

- The prime-color wavelengths correspond to the maxima of *tristimulus difference functions* $\bar{b} = \bar{z} - \bar{x} - \bar{y}$, $\bar{g} = \bar{y} - \bar{x} - \bar{z}$, $\bar{r} = \bar{x} - \bar{y} - \bar{z}$, which Thornton termed "bluminosity," "verdinosity," and "rubinosity." He proposed corresponding units (blumen, verden, ruben) analogous to the lumen, and showed that rubinosity ρ is an excellent predictor of red phosphor performance in tri-phosphor blends (Fig. 14 in the original).

Thornton's insight is that the prime-color wavelengths are not arbitrary engineering choices but reflect the structure of human color vision: they are the wavelengths that "pull hardest" toward their chromaticities per watt of power, and are therefore the most efficient at establishing a wide color gamut from a small number of spectral components.

6.2 Walter (1978): Nonlinear Programming of Lamp Spectra

Wolfgang Walter of GTE Sylvania [10] took a fundamentally different approach: rather than assuming a fixed number of spectral bands and optimizing their positions, he applied *nonlinear mathematical programming* (the generalized reduced gradient algorithm of Lasdon et al.) to find optimal spectra with *no a priori constraint on shape*. The spectrum was discretized into 31 variables (10 nm intervals from 400–700 nm), with the objective of maximizing luminous flux of a 40-watt fluorescent lamp subject to constraints on total UV excitation power, ANSI chromaticity ellipses, and a CRI floor.

Walter's key results:

- Optimal spectra with CRI up to 90–95 consist of **just four narrow bands**, positioned at approximately 460, 530, 580, and 620 nm, stable across all four ANSI lamp types (cool white, white, warm white, daylight).
- The unconstrained (CRI-free) optimum is a two-line spectrum (the MacAdam pair), confirming earlier theoretical work.
- As CRI constraints increase from 50 to 95, additional bands appear and existing bands broaden, with a monotonic decrease in luminous efficacy (from ∼140 lm/W at CRI 50 to ∼107 lm/W at CRI 95 for cool white).
- The optimization surface is "fairly flat" near the optimum: small perturbations to the SPD do not greatly affect luminance or CRI, suggesting practical tolerance for real phosphor emission profiles.
- When *all* component indices $R_j \geq C$ are required (not just the average R_a), efficacy drops further — confirming that the CRI average conceals weaknesses in individual color samples.

The published discussion section is notable: Koedam and Opstelten of Philips argued that the four-band optimum offers "no real improvement" over the Philips three-band lamp after correcting for realistic quantum efficiency (∼80% rather than 100%). Thornton himself commented, noting that if subjective criteria (color preference, visual clarity, perceived brightness) are considered rather than CRI alone, three bands may be superior to four.

6.3 Relationship to Our Results

Our `photopic` optimizer, approaching the problem from yet another direction — continuous piecewise-linear log-amplitude perturbation of a blackbody, optimized by COBYLA — independently converges to essentially the same spectral structure. The three-peak topology that dominates our results at low CRI floors (437 lm/W at CRI $\geq$ 60) corresponds to Thornton's prime colors, while the broadening and addition of a fourth peak at higher CRI floors (CRI $\geq$ 90 and above) matches Walter's four-band result. The wavelength positions we find (∼450, ∼540, ∼610 nm) agree with Thornton to within the resolution of our spectral parameterization.

The consistency across three independent methods — Thornton's parametric three-line analysis, Walter's unconstrained nonlinear programming, and our hierarchical COBYLA optimization — provides strong evidence that the prime-color structure is a genuine feature of the CRI/efficacy optimization landscape, not an artifact of any particular numerical method.

7 The Deficiencies of CRI and the Legislative Paradox

7.1 Why CRI is Known to be Wrong

The CIE Color Rendering Index (CRI, R_a) has been recognized as flawed since at least the 1980s, yet it remains the dominant regulatory metric. The principal deficiencies are:

1. **Only 8 unsaturated test samples.** R_a averages R_1 through R_8, all of which are low-chroma, pastel-like Munsell samples. Saturated colors — which are the ones most affected by narrow-band spectra — are represented only in R_9–R_{14}, which are reported but *not* included in the official score. This is the loophole our optimizer exploits: it can drive R_9 (saturated red) to -28 while maintaining $R_a = 65$.

2. **Von Kries adaptation model.** The CRI uses a simple diagonal (von Kries) chromatic adaptation transform, which is known to be a poor approximation of human color constancy, especially for narrow-band illuminants far from the blackbody locus.

3. **Uniform color space.** The CRI computes color shifts in the CIE 1964 $U^*V^*W^*$ space, which is perceptually nonuniform. Equal ΔE values do not correspond to equal perceived color differences.

4. **Averaging hides problems.** A lamp with seven $R_i = 95$ and one $R_i = 25$ gets $R_a = 86$ — an apparently acceptable score that conceals a severe deficiency in one color region. Walter's work showed that requiring all $R_j \geq C$ rather than just $R_a \geq C$ significantly changes the optimal spectrum and reduces achievable efficacy.

5. **Penalizes increased saturation.** Because CRI measures *fidelity* (closeness to the reference illuminant), a lamp that renders colors *more* saturated than the reference — which people often prefer — receives a *lower* CRI score. Thornton's color preference index (CPI) and the IES TM-30-15 standard [14] address this.

7.2 Why Legislators Use It Anyway

Despite these well-known deficiencies, CRI remains the basis for lighting regulations worldwide (EU Ecodesign 2019/2020, US DOE/EISA, California Title 24, Energy Star). The reasons are partly institutional inertia and partly the absence of a universally agreed replacement:

- **IES TM-30-15/18** defines two metrics: R_f (fidelity, similar to CRI but with 99 test samples and CAM02-UCS) and R_g (gamut area, measuring saturation changes). It is technically superior but has not yet been mandated by any major regulatory body.

- **CIE 224:2017** proposed a general color fidelity index R_f similar to TM-30, but adoption has been slow.

- The lighting industry has decades of CRI-referenced product data, specifications, and marketing materials. Changing the metric would require re-characterizing the entire installed base.

The result is a regulatory landscape that *incentivizes* the very spiky, CRI-gaming spectra that our optimizer finds. Manufacturers who maximize efficacy while just meeting CRI ≥ 80 are rewarded by the regulatory framework, even though the resulting light may render saturated colors poorly. The "weird final state" is not a theoretical curiosity — it is the market equilibrium toward which regulations push.

8 Future Work

Several directions suggest themselves:

1. **R_9 as a function of required efficacy.** The most practically important question is: under a given regulatory framework (e.g., EU Ecodesign minimum efficacy), what is the *achievable* R_9 (saturated red rendering)? Our current runs fix the CRI floor and let efficacy be the objective. The dual problem — fix efficacy and maximize R_9 — would map out the "quality frontier" that regulations actually define. Similarly, sweeping over required efficacy at fixed CRI floor would show how R_9 degrades as efficiency pressure increases.

2. **Realistic quantum efficiency.** Our calculations assume unit quantum efficiency, following Walter's approach. Real phosphors have wavelength-dependent quantum efficiencies $\eta_Q(\lambda)$, typically 80–95% for blue/green and dropping for deep red. Koedam and Opstelten noted that applying realistic η_Q reduces calculated efficacy by ∼20%. Incorporating $\eta_Q(\lambda)$ into the objective function would shift the optimal peak positions and show whether the three-peak structure survives under real-world conversion losses.

3. **TM-30 optimization.** Replace the CRI constraint with TM-30 R_f and R_g metrics. Because TM-30 uses 99 test samples and a more uniform color space, the optimizer would have far less room to exploit loopholes. The resulting spectra may be broader and more natural-looking.

4. **Multiple CCTs.** Extend the analysis to the full range of commercial color temperatures (2700–6500 K). Walter showed that the four-band positions are remarkably stable across ANSI lamp types; verifying this with our continuous parameterization would be valuable.

5. **LED-realistic spectra.** Constrain the spectrum to physically realizable LED spectra: a narrow blue pump peak (∼450 nm, 20 nm FWHM) plus one or more phosphor emission bands (50–100 nm FWHM, constrained shapes). This would bridge the gap between our theoretical optima and practical LED design.

6. **Scotopic/mesopic efficacy.** Outdoor and roadway lighting operates at mesopic light levels where the scotopic sensitivity function $V'(\lambda)$ contributes. The S/P ratio (scotopic to photopic luminance ratio) is an additional design variable that regulations are beginning to address.

7. **Circadian impact.** Blue-rich spectra suppress melatonin production. A constraint on melanopic equivalent daylight illuminance (M-EDI) could be added to the optimization, capturing the tension between visual efficacy and biological impact.

9 Conclusion

The constrained optimization of light spectra reveals that regulatory pressure on luminous efficacy drives lamp spectra toward a mathematically determined "weird" final state: three narrow spectral peaks near 450, 540, and 610 nm with suppressed emission elsewhere. This state maximizes lumens per watt while technically satisfying color rendering requirements, but at the cost of poor rendering of saturated colors — especially reds.

The `photopic` tool, built as a hybrid Modula-3/Scheme application, enables rapid exploration of this trade-off space using COBYLA constrained optimization with hierarchical spectral resolution.

Availability

The `photopic` tool, this report, the optimization data, and all plotting scripts are open source. The source code is available in two repositories:

- **Photopic tool and data:** `https://github.com/IntelLabs/async-toolkit`, branch `claude_edits`, path `m3utils/m3utils/photopic/`. The report and optimization results are in `photopic/report-runs/`.
- **CM3 compiler, MScheme, and runtime libraries:** `https://github.com/modula3/cm3`. MScheme lives in `m3-scheme/`; the COBYLA binding, spectral libraries, and CIE data are in the `m3utils` tree above.

Both repositories are licensed under open-source terms (Apache 2.0 for the async-toolkit; BSD for CM3).

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